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B.Tech. Degree IV Semester Examination April 2019

CE 402 (A/B) SURVEYING II (2006 Scheme)



Time: 3 Hours

Maximum Marks: 100

PART A

(Answer **ALL** questions)

(8 × 5 = 40)

- I. (a) Explain the elements of a compound curve with a neat sketch.
- (b) How will you obtain the length of a vertical curve?
- (c) Define triangulation and explain its classification.
- (d) State any five laws of weights.
- (e) Define the terms: latitude, longitude, altitude, azimuth and declination
- (f) Explain "The Equation of Time".
- (g) What are the methods of taking Sounding? Explain.
- (h) What are the applications of air photograph?

PART B

(4 × 15 = 60)

- II. (a) Two straight lines AT_1 and CT_2 meet at V. It is proposed to introduce a reverse curve of common radius R, having T_1 and T_2 as tangent points. The angles AT_1T_2 and VT_2T_1 measured at T_1 and T_2 are $45^\circ 30'$ and $25^\circ 30'$ respectively. The distance T_1T_2 is equal to 800 metres. Determine the common radius and central angle for two arcs. (7)
- (b) Describe briefly the procedure of setting out a circular curve by Rankine's method of tangential angles. (8)

OR

- III. (a) Explain the methods to find the Length of transition curves. (7)
- (b) Determine the radial offsets to be set out at half chain interval along the tangents to locate a 16-chain curve, the length of each chain being 20 m. (8)
- IV. (a) Write a short note on signals and towers. (5)
- (b) The altitudes of two proposed stations A and B, 100 km apart, are respectively 420 m and 700 m. The intervening obstruction situated at C, 70 km from A has an elevation of 478 m. Ascertain if A and B are intervisible, and, if necessary, find by how much B should be raised so that the line of sight must nowhere be less than 3 m above the surface of the ground. (10)

OR

- V. (a) State and explain the theory of least squares. (7)
- (b) Form the normal equations for x , y and z in the following equations: (8)

$$3x + 3y + z - 4 = 0, \quad \text{Weight} = 2$$

$$x + 2y + 2z - 6 = 0, \quad \text{Weight} = 3$$

$$5x + y + 4z - 21 = 0, \quad \text{Weight} = 1$$

(P.T.O.)

- VI. Find the zenith distance and the altitude at the upper culmination of the stars from the following data: (15)
- (i) Declination of star = $42^{\circ}15' N$ Latitude of observer = $26^{\circ}40' N$
 - (ii) Declination of star = $23^{\circ}20' N$ Latitude of observer = $26^{\circ}40' N$
 - (iii) Declination of star = $65^{\circ}40' N$ Latitude of observer = $26^{\circ}40' N$

OR

- VII. Calculate the sun's azimuth and Hour angle at the sunset at a place in latitude $42^{\circ}30' N$, when its declination is (i) $22^{\circ}12' N$ (ii) $22^{\circ}12' S$. (15)
- VIII. (a) Write a short note on Shore Line Survey. (5)
- (b) A, B and C are three visible stations in a hydrographical survey. The computed sides of the triangle ABC are: AB, 1130 m; BC, 1372 m; and CA, 1889 m. Outside this triangle (and nearer to AC), a station P is established and its position is to be found by three point resection on A, B and C, the angles APB and BPC being respectively $42^{\circ}35'$ and $54^{\circ}20'$. (10)

OR

- IX. (a) What is flight planning? Explain. (5)
- (b) The scale of an aerial photography is 1 cm = 100 m. The photograph size is 20 cm $\times$ 20 cm. Determine the number of photographs required to cover an area 10 km $\times$ 10 km, if the longitudinal lap is 60% and the side lap is 30%. (10)
